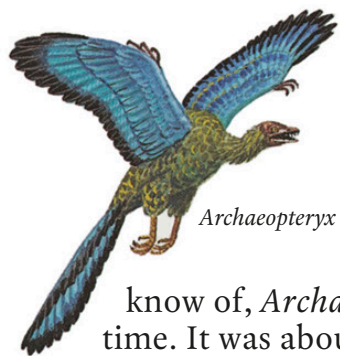


What Is a Bird?



Archaeopteryx

BIRDS ARE THE ONLY LIVING ANIMALS in the world that have feathers. They also lay eggs, breathe air, and are warm-blooded, meaning that they keep their body temperature the same at all times. Birds evolved from reptiles about 150 million years ago. The earliest bird we know of, *Archaeopteryx* (meaning “ancient wing”), lived at this time. It was about the size of a crow and had feathers, although it probably couldn’t fly very well. The 10,964 or so species of bird alive today have developed from early birds like *Archaeopteryx*.

A bird’s wing is light, strong, and flexible so it will not snap as the bird twists and turns through the air.



Bird bones

A bird has a bony skeleton inside its body to support and protect its delicate organs. But the whole skeleton of flying birds is very light and the long bones in their wings and legs are hollow, with a honeycomb of stiff supporting struts (above). This means they have less weight to lift off the ground and stay up in the air.

BIRD BILLS

Birds use their bills to catch and hold their food, care for their feathers, and build nests. The size and shape of a bird’s bill depends on what it eats and where it finds its food.

Pine grosbeak
(*Pinicola enucleator*)
Seed-eater

White-cheeked barbet
(*Megalaima viridis*)
Fruit- and insect-eater

African paradise flycatcher
(*Terpsiphone viridis*)
Insect-eater

Australian darter
(*Anhinga novaehollandiae*)
Fish-eater



Peregrine falcon
(*Falco peregrinus*)

A bird uses its tail feathers for steering as it flies through the air.

Birds have scaly legs like their reptile ancestors.



Body or contour feather from a red lory.

Flight feather from a Guinea fowl.

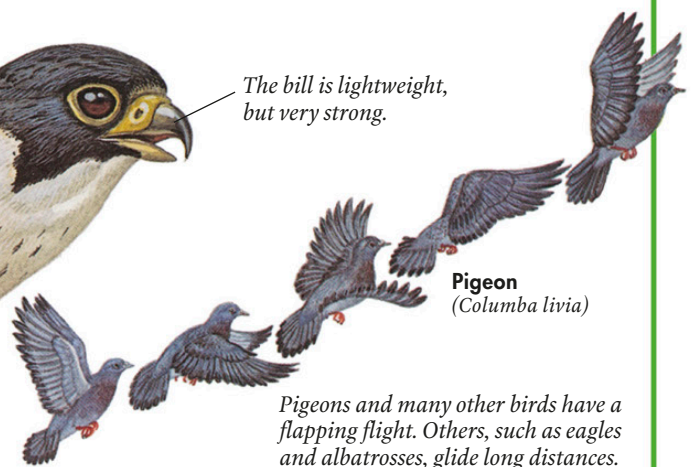
Down feather from a pigeon.



Feathers

A bird has a huge number of feathers. Even a small bird, such as a wren, has more than 1,000 feathers. There are three main types of feather—flight feathers on the wings and tail, body feathers which give a bird its shape, and fluffy down feathers to keep it warm. Flight feathers are made of strands called barbs which hook together. If the hooks come apart, they can be drawn together again, like zipping up a zipper. Each year, a bird sheds or molts most of its feathers and grows new ones to replace them.

The bill is lightweight, but very strong.



Pigeon
(*Columba livia*)

Pigeons and many other birds have a flapping flight. Others, such as eagles and albatrosses, glide long distances. Hummingbirds can even hover.

How birds fly

Birds are the largest, fastest, and most powerful flying animals alive. Their smooth, streamlined shape lets them cut through the air easily, while their powerful chest muscles flap their wings, pushing them along. A bird’s wings are curved on top and flat underneath. As a bird flies, this airfoil shape creates an area of high air pressure under the wing and an area of low air pressure above it. The high pressure under the wing pushes the bird up into the air. Some birds, such as penguins and ostriches, cannot fly—they run or swim very fast instead.

Eggs and chicks

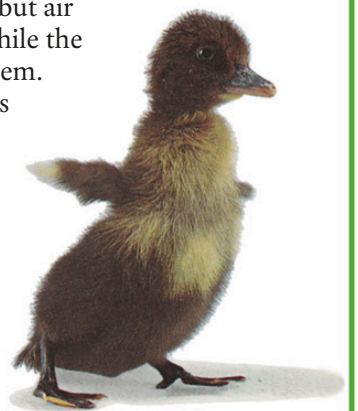
All birds lay eggs. The eggshell is hard to protect the developing chick, but air can still pass through. The chick also has a food store inside the egg. While the chick develops, the parents have to keep the eggs warm by sitting on them. This is called incubation. Many birds are blind, featherless, and helpless when they hatch. Other birds, such as this duckling, spend longer inside the egg and so are better developed when they hatch. They can run around and fend for themselves almost as soon as they hatch.



A duckling starts to break out of its shell by cutting a circle with its bill.



When the young bird pushes itself out of the shell its feathers are still wet.



Within three hours its feathers are dry and fluffy and it can run around.